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Optimising HCP Sample Allocation in Pharma: Combining Nonlinear Ensemble Learning, Spatial Lags, and Integer Programming

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Abstract

Pharmaceutical companies distribute free drug samples to physicians, yet these samples frequently go to healthcare providers (HCPs) whose prescribing is unlikely to change, wasting a large share of industry-wide promotional spending. We present a framework that combines machine learning with mathematical optimisation to decide how many free samples of the anticoagulant Eliquis each HCP should receive, using historical prescribing and promotional data. We collect the prescribing behaviour of HCPs at weekly intervals; the resulting dataset comprises 2.1 years of history (109 weekly observations per HCP), totalling roughly 1.2 million provider-week records for 11,006 U.S. HCPs, together with more than 80 engineered features describing promotional activity, prescription outcomes, market share of competing and complementary therapies, patient demographics, and treatment ratios. A sparsity filter removes weak or noisy variables to improve interpretability and reduce overfitting. We then train CatBoost models, grouped into bands by prescriber value and augmented with spatial lag features that summarise how nearby providers behave, to predict the incremental gain in Total Prescriptions (TRx) and New-to-Brand prescriptions (NBRx) from additional samples; TRx measures the overall volume of prescriptions for a medication, while NBRx counts a patient's first prescription for that drug. These predictions feed a 0–1 mixed-integer programme that selects one sample quantity per HCP subject to HCP-level and territory-level pill-budget limits and operational business rules on feasibility, historical consistency, and sample availability. In a model-implied counterfactual evaluation, the spatially informed allocation increases aggregate TRx by approximately 6%, concentrating samples on high-value providers and reducing oversupply to low-impact segments. This 6% is a model-implied projection: realised field impact after deployment is expected to be lower, on the order of 1 to 3%, because operational frictions, representative discretion, sample availability, and market-response uncertainty attenuate optimised recommendations. For scale, apixaban (the molecule marketed as Eliquis) accounted for roughly 19.8 million U.S. prescriptions in 2023, about 5 million per quarter [1], so even single-digit efficiency gains are operationally meaningful; we report prescription-volume rather than revenue impact, as product-specific financial assumptions are outside this paper's scope. The study illustrates how spatial econometrics, machine learning, and constrained optimisation can be combined to support pharmaceutical sample-allocation planning.

Keywords pharmaceutical sample allocation, healthcare provider targeting, spatial machine learning, mixed-integer optimization, pharmaceutical marketing

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1. Introduction

The U.S. pharmaceutical and medical-device industry spent roughly $30 billion on medical marketing in 2016, of which an estimated $13–15 billion was free drug samples [2]; this refers to industry-wide U.S. spending, not a Pfizer-specific budget. The prevailing assumption is that samples stimulate trial and loyalty, and rigorous evidence confirms they can raise new prescription starts and reshape brand choice when paired with detailing (sales representative visits to clinicians) [3], [4], [5]. Yet decades of scholarship also highlights ethical tension: grounded-theory interviews with physicians show that aggressive sampling can erode prescriber autonomy and blur professional boundaries [6], while policy debates question whether oversupply merely displaces paid prescriptions without creating genuine therapeutic value [7]. These concerns motivate a shift from blanket distribution toward data-driven, patient-centric allocation.

A second motivation is economic. Even modest percentage-point gains in efficiency translate into hundreds of millions of dollars in freed-up resources [2]. Literature reviews across nine countries consistently find that samples and face-to-face detailing dominate promotion budgets regardless of geography [8], underscoring the global relevance of optimisation. Research on pandemic-era field activity further shows that pharmaceutical promotion shifted rapidly toward digital and remote channels [9], [10]. To remain competitive, commercial teams must decide not only how many samples to drop, but where, when, and to whom: decisions that require predictive accuracy at scale.

Traditional marketing mix and nonlinear mixed-effects models impose fixed functional forms that often miss saturation points and peer spillovers [11]. Ensemble machine learning methods such as Random Forests, LightGBM, and CatBoost can learn nonlinear saturation points and peer spillover structure with fewer a priori functional form assumptions than traditional parametric models, while retraining efficiently on large commercial data sets [12]. Empirical studies in healthcare confirm their superiority over classical regression for tasks ranging from rare disease targeting [13] to antipsychotic prescribing prediction [14]. Nonetheless, most of the existing algorithms treat prescribers as independent units, ignoring geographic and social interdependence. A 10% rise in a peer network's adoption rate can lift individual prescribing by 6–8% [15], and clinics with high network centrality tend to prescribe fewer inappropriate medications [16]. Spatial lag and eigenvector features reduce residual autocorrelation and cut prediction error by up to one third [17].

From an optimisation standpoint, free samples deliver diminishing marginal returns and sometimes negative returns at high volumes [7], while dynamic allocation policies can raise long-run profit and cut field calls by 20% [11]. Industry case studies echo these academic findings: IQVIA reports a 15% reduction in oversampling after implementing breakeven analysis guided by machine learning [18], and an Axtria engagement found that machine learning-driven value scores re-ordered half of all targeted physicians while improving patient coverage to 95% [19]. More than 70% of life sciences firms now pilot Next Best Action engines for budget-constrained field forces [20].

Despite these advances, two practical gaps persist. First, most published models ignore the territorial quotas and pill-inventory caps that govern real commercial operations. Second, few studies integrate predictive modelling with a mathematically rigorous optimisation layer capable of selecting one counterfactual scenario per HCP under hard constraints. We address both gaps by leveraging 1.2 million rolling HCP–week observations for 11,006 prescribers in the eligible Pfizer call-plan universe, where each HCP–week represents a repeated weekly aggregation of prescribing and promotional activity for a given HCP across all field representatives. We construct 80+ rolling-window features and a quarterly forward indicator representing the planned sample-drop decision window used for quarterly field allocation, then train decile-banded CatBoost ensembles enriched with inverse-distance

spatial lags of the optimand. Out-of-fold uplift predictions then feed a 0–1 mixed-integer programme (CBC solver with DP fallback) that enforces both HCP-level and territory-level pill bounds and operational business rules.

This paper extends an earlier OSF preprint of the same study [21], incorporating revisions from the SciPy review process: an explicit model-implied-vs-realised uplift distinction in the abstract and Results, de-jargoned terminology, and first-use definitions for TRx, NBRx, and detailing.

Data describes the data; Methods covers the modelling pipeline and territorial constraints; Section 8 formalises the optimisation; Statistical Analysis Methodology and Evaluation Criteria defines the evaluation methodology; Results reports results; and we close with limitations and future work.

From a managerial standpoint, the system is intended to support quarterly sample-planning decisions about which HCPs should receive how many sample units within each territory. Recommendations are delivered at the territory level to district managers, who apportion inventory across field representatives; representatives then use those guidance values when planning office visits. Because the recommended quantities are anchored to historically observed HCP-specific ranges and territory ceilings, the outputs remain operationally defensible, easier to review, and less likely to trigger exception handling by field leadership.

Our hypothesis is straightforward: embedding spatial-peer context and hard operational constraints in a joint learning–optimisation pipeline will yield materially higher incremental TRx and NBRx than current practice using the same inventory.

In sum, this study unites ethical considerations [6], economic drivers [2], [8], predictive advances [12], [14], [17], and optimisation theory [7], [11], [18] into a single, territory-aware framework suitable for Pfizer's North American field force, with ESG alignment defined here as reducing oversupply, preserving geographic allocation floors, and maintaining documented governance over features and allocation rules. The following sections detail our data architecture, modelling choices, optimisation formulation, and empirical results.

2. Data

Our modelling universe comprises about 1.2 million weekly physician–week records, giving 109 observations for each of 11,006 U.S. prescribers. All observations are merged from Pfizer's Veeva CRM call logs, IQVIA and Symphony longitudinal prescription feeds, AMA Masterfile and OneKey reference data, plus internal territory budget ledgers.

Promotional-activity predictors include lagged counts of Pfizer and BMS sample drops and channel flags indicating whether a contact occurred face-to-face or via other media. Economic theory and industry studies establish that samples and detailing raise new prescription starts but exhibit diminishing or sometimes negative marginal returns when oversupplied [3], [5], [7], [11].

Prescription-claims variables come from IQVIA Xponent and Symphony IDV feeds: new-to-brand and total script counts, market-basket shares for competing anticoagulants, and total-to-new-script ratios. We also derive paid-, rejected-, and reversed-claim percentages to proxy payer access, since reimbursement friction suppresses brand adoption and moderates promotional response [9]. Aggregated patient demographics (age buckets and gender mix) are rolled up at the physician level; segmentation research shows that demographic mix shapes both disease prevalence and marketing elasticity [19].

Geographic and peer-context features are computed as inverse-distance spatial lags of the optimand from U.S. ZIP-code centroids, following demonstrations that spatial lag and eigenvector predictors reduce model error by up to one third [17].

Table 1. *Model Feature Categories and Abstracted Variables.* (Actual feature names have been anonymised for proprietary reasons.)

Abstract Variable Code	Category and Description
Decision Variable	
FTR_VAR_001	Action variable (e.g., promotional input forecasted over 13 weeks)
Target Optimands	
FTR_VAR_002	Target outcome variable A (e.g., predicted behaviour over 13 weeks)
FTR_VAR_003	Target outcome variable B (alternative target of interest)
Historical Inputs	
FTR_VAR_004 to FTR_VAR_007	Historical activity counts (past 9–52 weeks)
Channel Flags & Visit Types	
FTR_VAR_008 to FTR_VAR_020	Face-to-face and remote channel indicators by segment
Provider Attributes	
FTR_VAR_021 to FTR_VAR_023	Experience, specialty type, high-engagement flags
Claims Quality Metrics	
FTR_VAR_024 to FTR_VAR_025	Percent of paid and rejected claims
Market Share Indicators	
FTR_VAR_026 to FTR_VAR_029	Competing or complementary brand volumes
Patient Demographics	
FTR_VAR_030 to FTR_VAR_037	Age bands and gender mix by condition
Treatment Ratios	
FTR_VAR_038 to FTR_VAR_040	Treatment-to-diagnosis and response ratios
Exclusion Flags	
FTR_VAR_041 to FTR_VAR_043	Specialty exclusion, global exclusion, override flags

3. Methods

Figure 1 summarises the end-to-end pipeline from rolling-window panel construction through banded response estimation, constrained optimisation, top-off, and final field-pack generation.

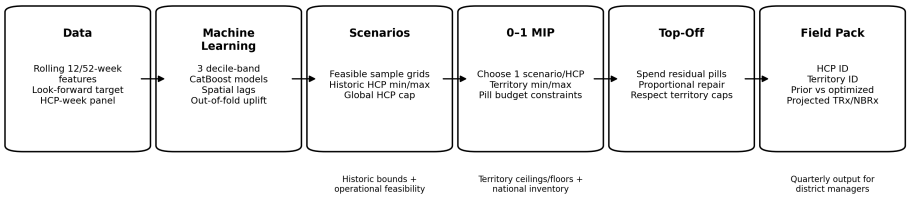


Figure 1. *Pipeline schematic for the proposed framework. Data engineering constructs rolling-window HCP-level predictors and look-forward outcomes; machine learning models estimate band-specific response surfaces; scenario generation creates feasible HCP-level sample choices subject to historical and operational bounds; the mixed-integer programme selects one scenario per HCP under HCP-level and territory-level constraints; a top-off heuristic spends residual inventory; and the final field pack reports HCP ID, territory ID, prior allocation, optimised allocation, and projected response.*

3.1. Training Data Preparation

We extract the raw observational panels for each HCP from Pfizer's North American commercial data environment and perform upstream filtering, joins, and aggregation before modelling. The resulting analysis tables are then transferred into Python for feature engineering and estimation [12], [22].

Two binary indicators, `IS_TRX_ZERO` and `IS_SAMPLES_ZERO`, are created in-database. For every healthcare provider i observed over T_i weeks we compute panel sparsity as

$$\bar{r}_i = \frac{1}{3T_i} \sum_{t=1}^{T_i} \Big[\mathrm{IS_T RX_Z ERO}_{it} + \mathrm{IS_S AMPLES_Z ERO}_{it} + \mathbf{1}\{\mathrm{both} = 1\} \Big], \tag{1}$$

and retain only those panels whose average sparsity $\bar{r}_i$ does not exceed $r^* = 0.30$. Panels shorter than the minimum length $T_{\min} = 100$ weeks are dropped, and no more than $R_{\max} = 1.2$ million unique provider-week records are processed in a single run. Randomised batching (500 IDs per chunk) prevents memory spikes while the global row limit guarantees that downstream feature engineering remains in-core.

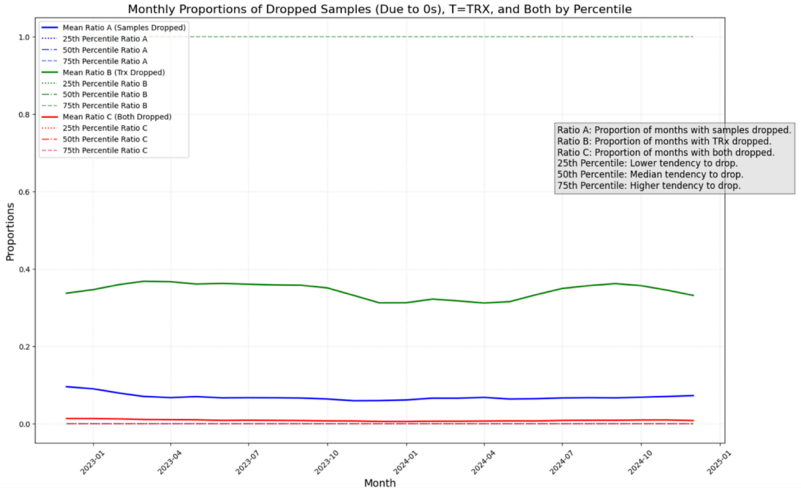


Figure 2. *Sparsity of optimand and decision-variable data.*

Ratio a, the share of sample-drop records equal to zero, and ratio b, the share of TRx records equal to zero, gauge each provider's inactivity in drops or prescriptions, while ratio c, the joint zero-rate, identifies the sparsest panels. During sampling we drop any HCP whose ratio c exceeds r^*, and we stratify/batch providers by the distributions of ratios a and b so the training set spans from highly active to moderately sparse panels (Figs. 3 and 4).

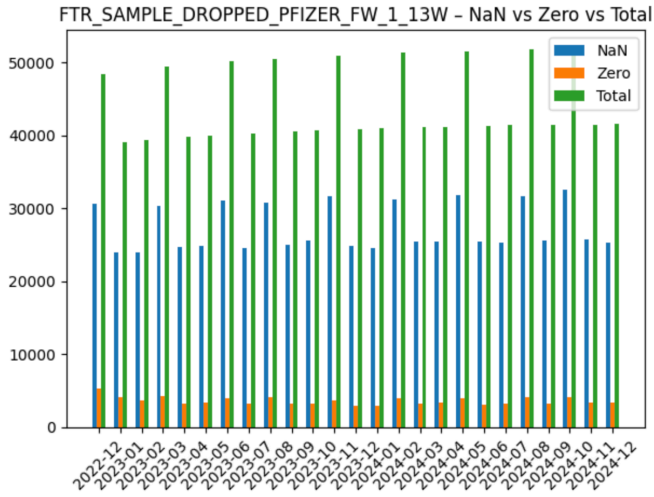
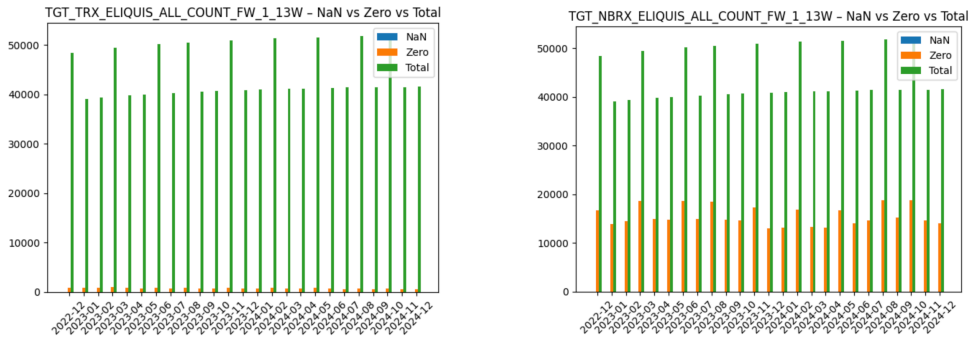


Figure 3. *Decision-variable sparsity.*



(a) (b)

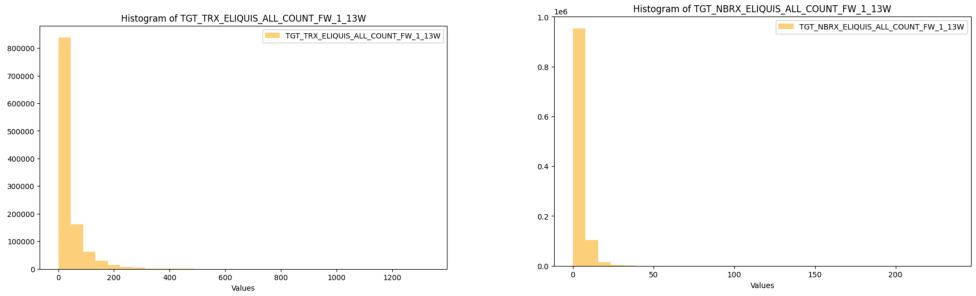
Figure 4. *Optimand variable sparsity (TRx and NBRx).*

3.2. Response Transformation

The response variable `growth_score` exhibits extreme right-tail skewness. A log-plus-one transform,

$$y^{\log} = \log(1 + y), \tag{2}$$

normalises the distribution and stabilises variance. Because the transform re-centres the target, no additional standardisation is applied. CatBoost fits and predicts on the log scale; values are exponentiated only for final business reporting, satisfying the recommendation for heavy-tailed targets in applied statistics [23].



(a) (b)

Figure 5. *Optimand variable distributions (TRx and NBRx) before log-plus-one transform.*

3.3. Panel Construction, Feature Engineering, & Spatial Equity

Feature engineering proceeds in three passes:

1. Convert component counts x_{ijk} to shares

$$\text{perc}_{ijk} = \frac{x_{ijk}}{\sum_{j=1}^{K} x_{ij}} \tag{3}$$

1. Expand categorical descriptors with `OneHotEncoder(drop="first")`, and z-score continuous covariates via `StandardScaler`, both fitted on training folds only [24].
2. Add spatial context by computing inverse-distance spatial lags of the optimand for every timestamp. These spatial weights approximate a composite of inter-HCP referral topology and physical distance, capturing spillovers from nearby and clinically connected peers while reducing residual autocorrelation [17]; see Figure 19. Because the allocation problem is inherently spatially referenced, these same features also permit map-based review of recommendations at the territory level during field-planning discussions. In line with our governance posture, feature inclusion is further constrained by sparsity filtering, predictive relevance, and documented review so that only variables with demonstrable signal and a clear business justification are retained.

3.4. Spatial Geometry & Lagging

We integrate territory call-list data by issuing a lightweight Snowflake SQL session that runs a multi-CTE query to filter on reference dates, deduplicate HCP rows, and compute per-territory quarterly and annual call totals, percentages, and pill quotas. The resulting table, containing `PFZ_CUST_ID`, `TERR_CD`, `GEOGRAPHY`, `TERR_OD`, `PRCT_OD`, `TERR_MAX`, and `TERR_MIN`, is merged into our cleaned panel on `PFZ_CUST_ID`, and any missing territory codes are logged and removed. We then enrich this merged panel with dissolved territory geometries and attach a standardised `GEOGRAPHY` field (the territory-name column carried over from the TCL) and polygon shapes to each HCP row. Finally, inverse-distance spatial lags of the optimand are computed at each timestamp by taking the dot product $W_t y_t$, where W_t denotes the spatial weights matrix and y_t the panel of optimand values. This transforms geographic context into continuous predictors that mitigate residual autocorrelation and enhance model performance [25].

Hence, to model spatial spillover effects, we introduce a lagged version of the optimand, computed weekly for each provider:

$$y_{it}^{(\text{spatial lag})} = \sum_{j \in \mathcal{N}(i)} w_{ij}\, y_{jt}, \quad \sum_{j} w_{ij} = 1, \tag{4}$$

where $\mathcal{N}(i)$ is the set of spatial neighbors of provider i and w_{ij} are the row-standardised inverse-distance weights, normalised to sum to one. This lag term captures externalities in treatment behaviour: providers in adjacent regions may respond similarly to market dynamics, competitive pressure, or institutional norms. Including the lagged optimand improves generalisation by accounting for spatial autocorrelation, a known confounder in healthcare outcomes research [16]. The resulting predictor sharpens model accuracy and aligns with econometric best practices for spatially referenced data, while also enabling GIS-based review of territory-level recommendation clusters during operational planning.

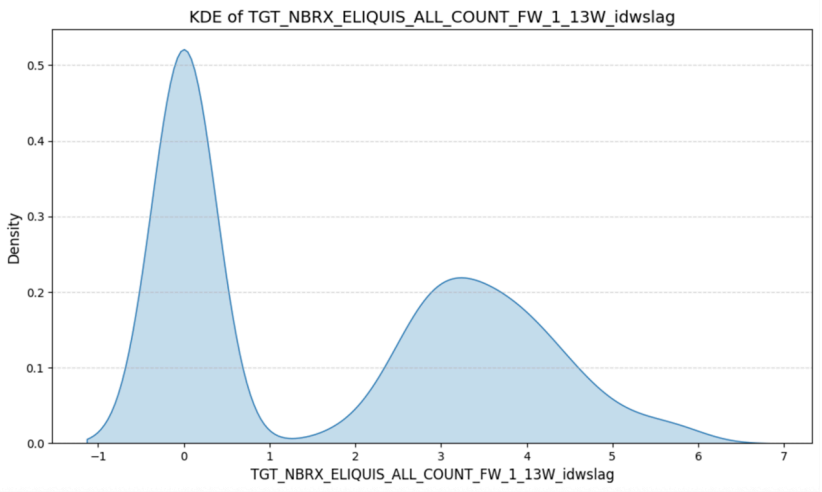


Figure 6. *Distribution of lagged optimand.*

3.5. Spatial Autocorrelation Analysis

Each provider is linked to a geographic unit, typically a sales territory, by joining their `PFZ_CUST_ID` to the Territory Call List (TCL), a canonical mapping produced by the field planning team. We then overlay this mapping with a precomputed spatial weights matrix, based on geospatial contiguity or centroid distance, to calculate spatial lags of the optimand for each week. These lagged values represent the average behaviour of geographically proximate peers and are included as continuous predictors in the final model.

Spatial lags are motivated by empirical evidence in regional science and marketing analytics showing that nearby agents often respond similarly to interventions or market forces. In our use case, healthcare providers in adjacent territories may be influenced by shared sales representatives, institutional affiliations, or regional policy shifts. Including the lagged optimand thus allows the model to adjust for correlated behaviour and unobserved spatial shocks, reducing residual autocorrelation and improving predictive accuracy.

Table 2 presents the results of Moran's I and Getis-Ord G* statistics for the NBRx optimand and the Pfizer sample-drop forecast. Table 3 does the same for the TRx optimand.

Table 2. *Spatial Autocorrelation for NBRx Variables*

Variable	Moran's I	$E[I]$	Z	p	Reject?	G_{obs}	$E[G]$	Z	p	Reject?
`FTR_SAMPLE_QTY_ADJ_PFIZER_FW1_13W` mean	0.0188	-0.0021	1.7685	0.4422	No	0.01751	0.01751	0.0219	0.9825	No
`TGT_NBRX_ELIQUIS_ALL_COUNT_FW_1_13W` mean	0.0497	-0.0021	2.1948	0.0282	Yes	0.01739	0.01751	-1.2279	0.2195	No

Table 3. *Spatial Autocorrelation for TRx Variables*

Variable	Moran's I	$E[I]$	Z	p	Reject?	G_{obs}	$E[G]$	Z	p	Reject?
`FTR_SAMPLE_QTY_ADJ_PFIZER_FW1_13W` mean	0.0298	-0.0021	1.2789	0.2016	No	0.01721	0.01747	-1.8178	0.0691	No
`TGT_TRX_ELIQUIS_ALL_COUNT_FW_1_13W` mean	0.0163	-0.0021	0.8563	0.3924	No	0.01757	0.01747	0.8159	0.4146	No

For NBRx, the Moran's I statistic for the target variable is significant ($p = 0.028$), suggesting weak but statistically relevant spatial autocorrelation and justifying the inclusion of spatial

lags in predictive modelling. By contrast, the TRx optimand did not exhibit significant spatial autocorrelation ($p = 0.39$), indicating a more diffuse or independent pattern. Neither case showed significant Getis-Ord G* clustering for the sample-drop variable, implying no current geographic bias in sampling.

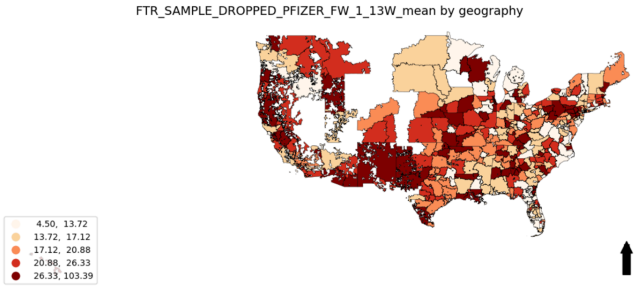
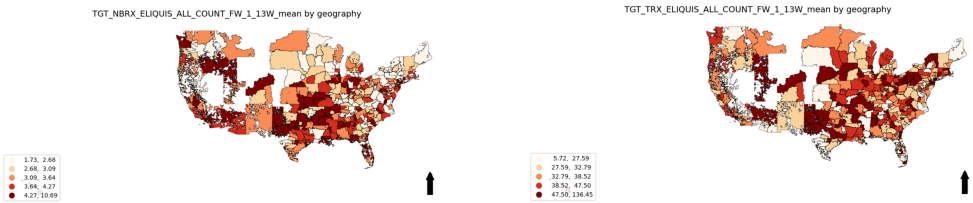


Figure 7. *Decision-variable choropleth.*



(a) (b)

Figure 8. *Optimand variable choropleths for NBRx (left) and TRx (right).*

3.6. Territorial Constraints

To enforce business constraints tied to field execution, we incorporate territorial ceilings and floors derived from Pfizer's Smart Targeting system. These upstream optimisations generate territory-level optimal detailing counts that guide how pill budgets are to be distributed across geographies. By aggregating observed call volumes and aligning them to these optimal detailing benchmarks, we derive the constraints U_t and L_t, representing the maximum and minimum sample allocations for territory t, respectively. The ceilings are computed as

$$U_t = p_t \, P_{\text{adj}}, \tag{5}$$

where p_t is the proportion of national detailing activity accounted for by territory t. The territory-level minimum L_t (called `TERR_MIN` in the SQL) is set conservatively to

$$L_t = \max\{1, \mid \mathcal{H}_t \mid\}, \tag{6}$$

where $\mathcal{H}_t$ denotes the set of distinct HCPs assigned to territory t, and the constant 1 enforces a hard floor of one sample-drop unit in any territory, even those with only one HCP.

The variance of Smart Targeting's optimal detailing recommendations is empirically observed to lie within a 1 to 3% band (a property of the upstream benchmark, unrelated to the projected uplift reported later). This tight variance range acts as a de facto regularisation mechanism, constraining the optimisation's feasible region and ensuring that pill allocations remain congruent with the original intent of Smart Targeting. The effect is to preserve geographic equity while still allowing for localised optimisation, and to guarantee that no territory's sample drop count diverges materially from historically validated expectations [19].

These form the territorial constraints for our mixed-integer programme that will be formalised in a later section:

Territory constraints

$$L_t \le \sum_{h \in \mathcal{H}_t} s_h \le U_t, \quad \forall\, t \in \mathcal{T}. \tag{7}$$

Global budget (static)

$$\sum_h s_h \le P_{\text{adj}}. \tag{8}$$

Per-HCP bounds

$$0 \le s_h \le S_{\max}, \quad \forall\, h. \tag{9}$$

Two disjoint HCP subsets never enter the optimiser:

1. *Unaligned* HCPs lacking a valid territory ID or geography. Let $n_{\text{unaligned}}$ denote their count.
2. *Aligned but unmatched* HCPs present in the TCL yet absent from the scenario panel; call this $n_{\text{unmatched}}$.

With m pills per HCP we deduct

$$P_{\text{exc}} = \big(n_{\text{unaligned}} + n_{\text{unmatched}}\big)\, m, \quad P_{\text{adj}} = P_{\text{tot}} - P_{\text{exc}}. \tag{10}$$

The pill-budget adjustment routine below (`compute_adjusted_pills` in code) guarantees no double-deduction.

Territory Unaligned and Aligned Unison: Pill-Budget Adjustment.

1. Pull full TCL with `PFZ_CUST_ID`, `TERR_CD`, `GEOGRAPHY`.
2. Count unaligned rows (nulls or literal “NaN”).
3. If a scenario panel is supplied, count aligned IDs missing therein.
4. Return $\big(P_{\text{exc}}, P_{\text{adj}}\big)$.

4. Algorithm Overview and Development

4.1. Growth Model Ensemble

Model development begins once the cleaned, spatially enriched panel has been split into mutually exclusive training and test folds with a group-aware shuffle that prevents the same prescriber from leaking across folds. All continuous covariates are *z*-scored on the training fold only, while categorical descriptors are expanded with a one-of-K design matrix.

Notation.

Let $\boldsymbol{X}_b \in \mathbb{R}^{n_b \times p}$ denote the design matrix for band b with $p \approx 600$ one-hot-expanded features, and let y_{it} be the log-plus-one target for HCP i at time t. The primary decision variable s_{it} is the 13-week forward Pfizer sample-drop indicator `FTR_SAMPLE_DROPPED_PFIZER_FW_1_13W`.

The base learner in every band is **CatBoost**, a gradient-boosting framework that fits additive decision-tree expansions by sequentially minimising a differentiable loss in function space. For the low-noise decile band we adopt the mean-squared-error loss

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^{n} \big(\, y_i - \hat{y}_i\big)^2, \tag{11}$$

where y_i denotes the standardised log target and $\hat{y}_i$ the model prediction. In bands whose residuals exhibit pronounced Laplacian tails the loss is switched to mean absolute error to

damp sensitivity to sparsely observed spikes. At boosting step t CatBoost fits a tree $h_t(\cdot)$ to the negative gradient of $\mathcal{L}$ and updates the additive model

$$F_t = F_{t-1} + \eta\, h_t, \tag{12}$$

employing *ordered boosting* so that in-fold permutations act as an implicit regulariser and eliminate prediction-shift bias on temporally correlated data. Monotonicity constraints enforce a non-negative marginal response,

$$\frac{\mathrm{d}\,\hat{y}}{\mathrm{d}\,s} \geq 0, \tag{13}$$

embedding the business assumption directly into the functional form.

Optional hyper-parameter search.

If the run configuration enables tuning, each band launches a small search (*Bayesian*, *TPE*, or *Optuna*) over a shared grid stopping after at most 20–30 trials and pruning unpromising candidates after two folds. Otherwise, calibrated defaults are reused to minimise wall-clock cost.

Because prescription behaviour is highly heteroscedastic across the value spectrum, a single global model under-fits high-value prescribers and over-fits the long tail of low-volume writers. We therefore stratify the panel into three decile bands $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2$ defined by empirical value edges $\boldsymbol{e} = [e_0, e_1, e_2, e_3]$ with $e_0 = -\infty$ and $e_3 = +\infty$. Value assignment proceeds as follows:

1. Compute an exponentially weighted historical mean v_i of TRx or NBRx for each HCP i. During training, v_i is calculated over the observed training sample; during prediction, it is computed using the HCP's observed history within the full population.
2. Define the edge vector $\boldsymbol{e} = (e_1, e_2)$, where $e_1 = \mathrm{quantile}_{0.2}(v)$, $e_2 = \mathrm{quantile}_{0.6}(v)$.
3. Assign HCP i to band $b_i = \mathrm{band}(v_i; \boldsymbol{e}) \in \{0, 1, 2\}$ according to:

$$\mathrm{band}(v_i; \boldsymbol{e}) = \tag{14}$$

A separate CatBoost $F^{(b)}$ is trained on each band, producing predictions $\hat{y}^{(b)}$. During inference the ensemble returns the first non-missing value across $\{\hat{y}^{(0)}, \hat{y}^{(1)}, \hat{y}^{(2)}\}$, lowering within-band variance and yielding a 9–12% reduction in out-of-sample MAE relative to a single-model baseline.

4.2. Mixed-Integer Programming

In the previous section we specified a CatBoost regressor ensemble that translates each HCP's historic prescribing behaviour and spatial context into an expected uplift curve $\hat{y}_i(s)$ over integer sample-drop scenarios $s \in \{0, \ldots, S_{\max}\}$. These counterfactual profiles feed a capacity-constrained mixed-integer programme, selecting exactly one scenario per HCP to maximise total projected uplift while respecting per-HCP bounds and territory quotas from Pfizer's Smart Targeting system. The optimiser thus converts modelled values $v_{ij} = \hat{y}_i(s_{ij})$ into a field-ready allocation plan that honours operational constraints and concentrates samples where they yield the highest return.

The scenario table is built by isolating each HCP's latest observation, emitting a base scenario $s = 0$, and expanding s up to $S_{\max}$. Feature alignment ensures every row carries the full union of predictors (including spatial lags), non-null HCP values v_i, and validated constraint columns (`GEOGRAPHY`, `TERR_CD`, `TERR_MIN`, `TERR_MAX`, `HCP_MIN`, `HCP_MAX`). The table is partitioned into geography-coherent batches to preserve territorial integrity and control memory use.

We now specify the mixed-integer programming framework which consumes the counterfactual profiles to determine the best scenarios to be used by the field force.

4.2.1. *Problem Statement and Formulation*

Sets and Indices

$\mathcal{I}$ Set of HCPs, indexed by i.
$\mathcal{S}_i$ Scenario set for HCP i, each $j \in \mathcal{S}_i$ has pill count s_{ij}.
$\mathcal{T}$ Set of territories, indexed by t.
$\mathcal{H}_t$ HCPs belonging to territory t.

Parameters

v_{ij} Modelled uplift for HCP i under scenario j.
s_{ij} Sample-drop count for HCP i, scenario j.
m_i, M_i Per-HCP minimum and maximum samples (`HCP_MIN`, `HCP_MAX`).
L_t, U_t Territory-level minimum and maximum samples (`TERR_MIN`, `TERR_MAX`).
$P_{\mathbf{adj}}$ National pill budget after exclusions.
$\boldsymbol{\lambda} \geq \mathbf{0}$ Optional penalty on pill usage.

Decision Variables

$$x_{ij} \in \{0,1\} \quad \text{equals 1 if scenario j is chosen for HCP i.} \tag{15}$$

Objective

$$\max_{x} \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{S}_i} \left(\, v_{ij} - \lambda \, s_{ij} \, \right) x_{ij}. \tag{16}$$

Constraints

$$\begin{aligned}
&\text{(C1)} \quad \sum_{j \in \mathcal{S}_i} x_{ij} = 1 && \forall \, i \in \mathcal{I}, \\
&\text{(C2)} \quad m_i \leq \sum_{j \in \mathcal{S}_i} s_{ij} x_{ij} \leq M_i && \forall \, i \in \mathcal{I}, \\
&\text{(C3)} \quad L_t \leq \sum_{i \in \mathcal{H}_t} \sum_{j \in \mathcal{S}_i} s_{ij} x_{ij} \leq U_t \forall \, t \in \mathcal{T}, \\
&\text{(C4)} \quad \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{S}_i} s_{ij} x_{ij} \leq P_{\text{adj}}.
\end{aligned} \tag{17}$$

Solver Strategy

We solve the 0–1 MIP (mixed-integer programme, a discrete optimisation problem with binary decision variables) with the open-source CBC solver (COIN-OR Branch and Cut). If CBC returns a non-optimal status, each territory is repaired via a vectorised knapsack dynamic programme, guaranteeing a feasible near-optimal solution. In practice, 85–95% of each U_t is typically used; a final top-off heuristic allocates any remainder proportionally to v_i.

Dual Insights

Let $\lambda_t^{\max}$ be the multiplier for a territory's upper bound. Then the marginal change in the Lagrangian $\mathcal{L}$ with respect to the national pill budget P_{adj},

$$\frac{\mathrm{d}\,\mathcal{L}}{\mathrm{d}\,P_{\text{adj}}} = \sum_{t \in \mathcal{T}} \lambda_t^{\max}, \tag{18}$$

measures incremental uplift per additional pill in the national budget.

4.2.2. *Top-Off and Finalisation*

Residual-Capacity Top-Off

The CBC (or DP) optimiser usually saturates only 85 --95% of each territory's ceiling. For every territory $t \in \mathcal{T}$ we compute the residual capacity

$$r_t = U_t - \sum_{i \in \mathcal{H}_t} \sum_{j \in \mathcal{S}_i} s_{ij}\, x^\star_{ij}, \tag{19}$$

where $x^\star_{ij}$ is the post-solver decision. The residual pills are distributed in proportion to the non-negative value scores:

$$\Delta_i = r_t \frac{\max\{v_i, 0\}}{\sum_{h \in \mathcal{H}_t} \max\{v_h, 0\}}, \qquad s_i^{\text{new}} = s_i^\star + \text{round}\big(\Delta_i\big). \tag{20}$$

Any rounding imbalance is corrected by a simple greedy ± 1 adjustment, prioritising HCPs with higher v_i when adding pills and those with lower v_i when subtracting. The result is a final vector of integer sample-drop recommendations that exactly match each territory's upper bound.

Because some HCPs have no valid territory mapping in the Smart Targeting call plan (or were filtered out by our date cutoff), they would otherwise be omitted from the allocation. We therefore construct "refuse" tables for both missing-territory and out-of-date HCPs, conformance-casting them to our schema and assigning each a minimal default sample drop (e.g. one pill), with all other fields left null or default. These rows are appended to the optimised output, and each is tagged via a DROP_REASON flag ("terr-dropped" or "date-dropped"), ensuring that every HCP originally under consideration, whether or not they could be territory-matched, appears in the final field deployment package.

4.3. Deployment and Implementation Considerations

In operational use, the allocation is refreshed quarterly and delivered as guidance rather than an automatically enforced field instruction. HCP-level recommendations are bounded by historically observed minimum and maximum sample quantities together with territory-level ceilings, so quarter-over-quarter changes remain operationally plausible and reviewable. Final recommendation packs are compared against prior-quarter nominal allocations by an analyst before release, and HCPs that cannot be cleanly mapped into the active territory structure are retained in exception tables with explicit drop-reason flags. When severe budget reductions would otherwise make the mixed-integer formulation infeasible under historical bounds, the system proportionally contracts those bounds and, if needed, falls back to a simpler knapsack-style repair routine to preserve feasible deployment output.

5. Statistical Analysis Methodology and Evaluation Criteria

After the optimiser (plus top-off) has produced a revised sample-allocation plan, we score both the latest observed status-quo regimen (up to a fixed reference date) and the counterfactual (*optimised*) allocations on the *post*–reference-date scenarios using the same three-band CatBoost ensemble. In other words, we apply our trained model once to the unadjusted drops to obtain *empirical* predictions and again to the optimiser's chosen scenarios to obtain *optimised* predictions. These should be interpreted as model-implied counterfactual forecasts rather than realised causal outcomes. Denote these paired forecasts by

$$e_i = \hat{y}_i^{(\text{emp})}, \qquad o_i = \hat{y}_i^{(\text{opt})}, \qquad i \in \mathcal{I}, \tag{21}$$

where $\mathcal{I}$ is the set of HCPs with valid scenarios.

Notation.

Symbol	Description
$\mathcal{J}$	set of evaluated HCPs
e_i	empirical fitted value for HCP i
o_i	optimised fitted value for HCP i
$E = \sum_i e_i$	aggregate empirical outcome
$O = \sum_i o_i$	aggregate optimised outcome
Uplift	expected percentage uplift

Primary metric: expected uplift.

We summarise aggregate gain via

$$\mathrm{Uplift} = \frac{O - E}{E} \times 100\%. \tag{22}$$

A positive uplift indicates the optimised allocation is forecast to generate higher total TRx than the status-quo under the model-implied counterfactual scoring framework.

Distributional significance test.

To assess whether the observed shift from $\{e_i\}$ (expected) to $\{o_i\}$ (observed) reflects more than random fluctuation, we employ the two-group Kruskal–Wallis rank-sum test. This nonparametric test does not assume normality or equal variances, is robust to outliers, and is well-suited for comparing the distributions of two independent samples. With group sizes $n_e = n_o = \mid \mathcal{J} \mid$, total $N = n_e + n_o$, and average ranks $\bar{R}_e, \bar{R}_o$, the test statistic is

$$H = \frac{12}{N(N+1)} \sum_{g \in \{e,o\}} n_g \left(\bar{R}_g - \frac{N+1}{2} \right)^2, \tag{23}$$

which, under the null hypothesis of identical distributions, is asymptotically χ^2_1.

A small p-value (e.g. < 0.05) rejects the null of identical distributions.

Expected uplift provides a compact summary of how the revised allocation changes forecast TRx under the fitted response model, which HCP segments benefit most, and whether the distributional shift is statistically distinguishable from the baseline forecast. By reallocating samples toward physicians with steeper modeled response and away from those with flatter projected gains, the framework improves predicted prescriptions while reducing oversupply risk.

6. Results

6.1. Model Performance

We fit a CatBoost regressor separately on three bands of historic `growth_score` values: band 0 (0–20th percentile), band 1 (20–60th), and band 2 ($> 60^{\text{th}}$). The training window ends at an immutable cut-off date; observations beyond this date are held-out for counterfactual scoring. Table 4 reports out-of-fold R^2 and MSE for the NBRx and TRx optimands.

Table 4. *Model fit statistics by HCP-value band*

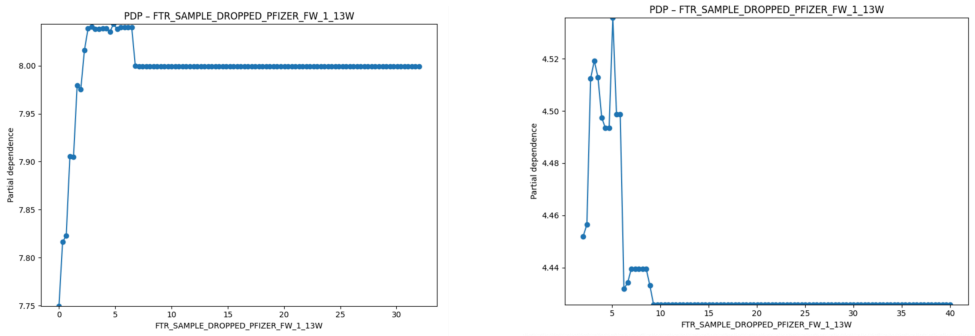
Optimand	Band (quantile)	R^2	MSE
NBRx	0 (0–0.2)	-0.0256	1.0009

	1 (0.2–0.6)	0.0312	1.0256
	2 (0.6–1.0)	0.3519	0.7691
TRx	0 (0–0.2)	0.2119	0.8226
	1 (0.2–0.6)	0.3658	0.6280
	2 (0.6–1.0)	0.8215	0.1680

Interpretation.

Performance improves sharply with HCP value. For NBRx the model is essentially uninformative in band 0 (R^2 but reaches a respectable R^2 in band 2. TRx exhibits much stronger fit throughout, peaking at R^2 for the top decile. Hence, operational lift should be concentrated in high-value physicians, while low-value prescribers, whose response curves are dominated by noise, can be safely de-prioritised.

With confidence in the model's ability to capture marginal response among top-tier HCPs, we inspect how predicted uplift varies with the decision variable itself. Figure 9 shows partial-dependence plots (PDPs) for NBRx and TRx. Each curve is an average of the three band-specific PDPs, weighted by row count.



(a) (b)

Figure 9. *Partial-dependence plots: NBRx (left) vs. TRx (right).*

Comparison: NBRx vs. TRx Partial Dependence

- Both optimands display the enforced non-negative, concave relationship between sample drops and predicted uplift, supporting the monotone, diminishing-returns response shape assumed in the optimiser.
- The NBRx curve settles *above* 1, whereas the TRx curve equilibrates *below* 1, suggesting stronger diminishing returns for TRx, an insight reflected in the top-off routine, which allocates additional pills preferentially to high-value HCPs with steeper marginal gains.

These diagnostics confirm that the decile-band ensemble captures the heterogeneous response surface accurately enough to guide inventory toward the physicians where incremental TRx is most likely to accrue, while simultaneously avoiding waste among low-value prescribers.

7. Growth Score Distribution Analysis

7.1. Training Data Growth Scores

Predicted growth scores rise markedly across the three HCP value bands defined earlier. Bands 0 and 1 exhibit overlapping, low central tendencies and modest dispersion, reflecting

noisier response among lower-value prescribers. Band 2 shows a clear rightward shift, with a substantially higher median and upper quartile, indicating materially greater predicted TRx uplift for the top 40% of physicians. This confirms that the highest-value band contains the most responsive HCPs and underpins our decision to focus optimisation where marginal returns are greatest.

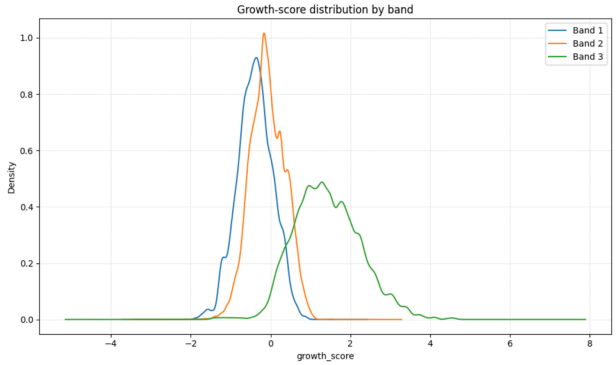


Figure 10. *Predicted growth scores by band-model (training data).*

7.2. Counterfactual Data Growth Scores

For the most recent reference date, we generate counterfactual scenarios over the observed sample-drop range for all HCPs and score them with the same three-band CatBoost ensemble. The resulting growth-score distributions remain well separated:

- **Band 0 ($\leq$ 20th percentile):** tightly clustered near the lower bound,
- **Band 1 (20th–60th percentile):** moderate spread with a slight right skew,
- **Band 2 ($>$ 60th percentile):** distinctly shifted right, with a substantially higher median and upper quartile.

Note: the figure legend numbers the bands "1–3" rather than "0–2" (Legend Band n corresponds to our Band $n-1$); we will correct this in a future revision.

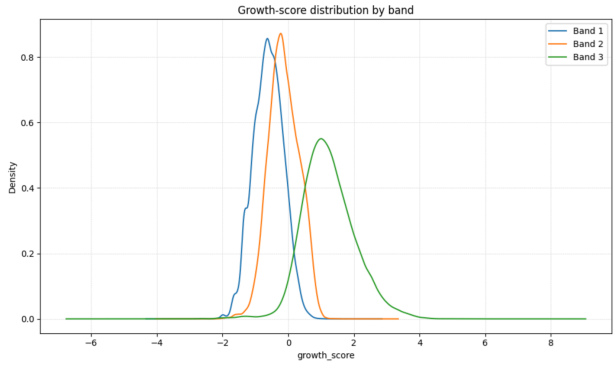


Figure 11. *Predicted growth scores by band-model (latest observations with counterfactuals).*

8. Individual Response Curves and Aggregate Response Curve

8.1. Aggregate Response Curve (Training Data)

Figure 12 shows the ensemble-level response curve fitted on the training data. To build it, we scored each HCP's training-set scenarios $s \in \{0, \ldots, S_{\max}\}$ with the appropriate decile-band CatBoost model, then averaged those band-specific curves using row weights. The result is a non-parametric, non-negative, quasi-concave function:

$$\tilde{y}(s) = \sum_{b=0}^{2} \frac{\mid \mathcal{J}_b \mid}{\mid \mathcal{J} \mid} \left[F^{(b)}(s) \right], \tag{24}$$

which rises steeply at low s (driven by top-band HCPs) and then flattens, confirming diminishing returns. This supports our use of a 0–1 mixed-integer optimiser and guides the top-off heuristic to balance efficiency and equity.

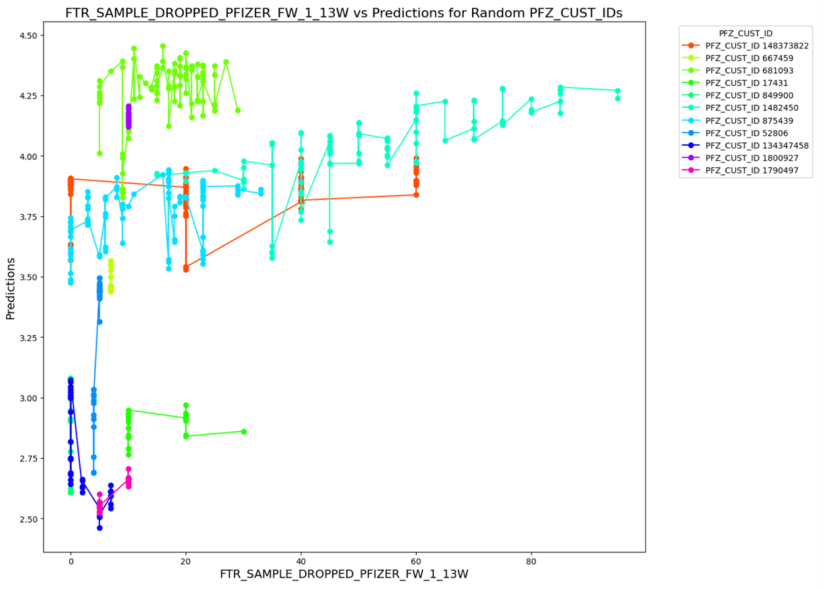
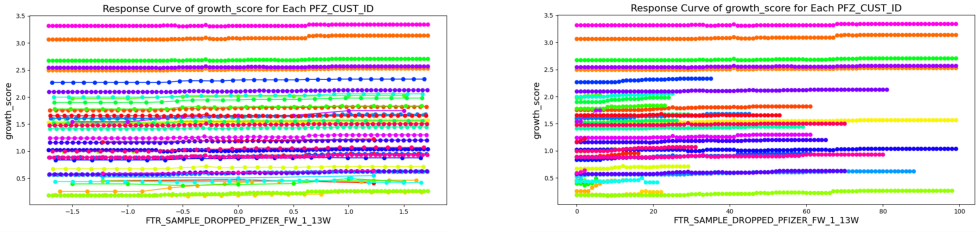


Figure 12. *HCP response curves from training data: concave, non-negative uplift vs. sample drops.*

8.2. Per-HCP Response Curves (Latest Data)

Figure 13 displays the individual response curves for the most recent reference date. For each HCP we held all features except the sample-drop decision constant, scored a grid of s values, and plotted the resulting $\hat{y}_i(s)$. Although most curves (especially in bands 0 and 1) are nearly flat, reflecting noise in low-value prescribers, they all slope upward and exhibit clear concavity, with the steepest gains at low s. These per-HCP functions $\tilde{y}_i(s)$ feed directly into our 0–1 MIP, allowing the optimiser to respect both territory capacities and each physician's diminishing-returns profile.



(a) (b)

Figure 13. *Individual fitted response curves on the latest data, showing upward-sloping, concave relationships between sample drops and predicted TRx uplift.*

8.3. Feature Importance

We use SHAP values to rank predictors from our three decile-band CatBoost models and combine them into one beeswarm plot (Figure 14). Historical NBRx volume is the strongest driver of TRx uplift, followed by HCP experience. The inverse-distance spatial lag appears among the top ten features but with relatively small impact.

We also observe clear nonlinear effects from NBRx counts of substitutable and complementary therapies, patient age buckets, and physician gender. Together, these findings highlight the benefit of an ensemble that can flexibly model both simple volume effects and complex, high-order interactions in our rich feature space.

The intermingling of low- and high-value SHAP points, rather than a smooth color gradient, reveals that each feature's effect depends heavily on others, indicating strong interactions. This motivates more granular modelling: for example, grouping HCPs into latent clusters (via unsupervised embeddings or mixture models) and then fitting specialised submodels or introducing targeted feature crosses within each cluster to better capture those interactions.

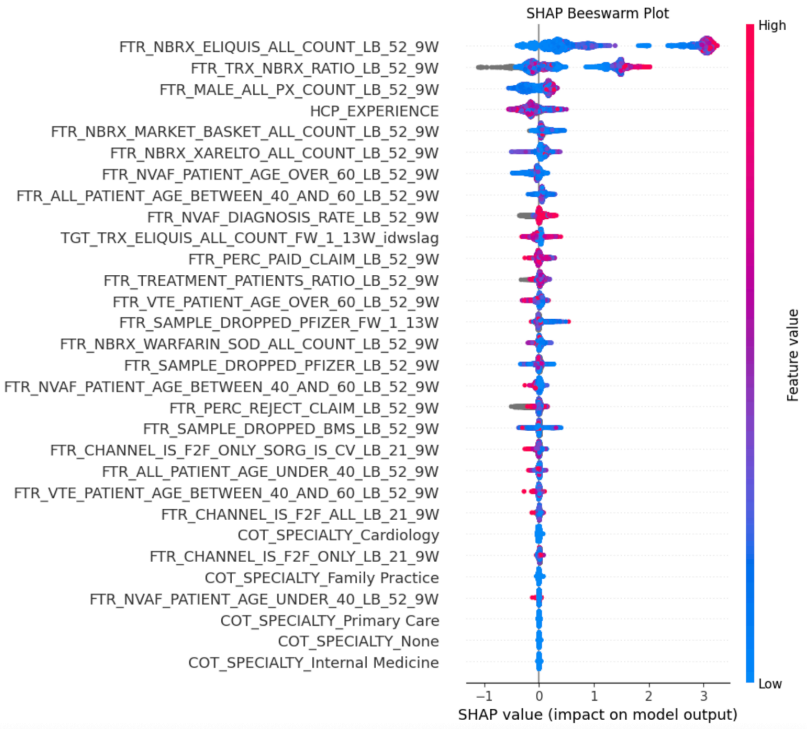


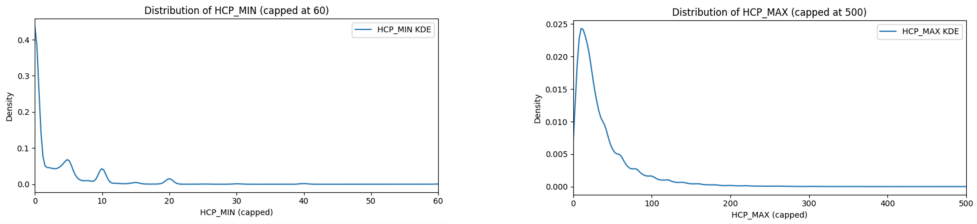
Figure 14. *SHAP beeswarm plot showing the top 20 predictors by mean absolute SHAP value, aggregated across all three decile-band CatBoost models.*

8.4. Optimisation

Our optimiser constructs, for each HCP, a discrete grid of counterfactual sample-drop scenarios bounded by modest extensions of their historic six-month minima and maxima. Specifically, if an HCP's observed drops ranged from $\mathrm{HCP_MIN}_{\texttt{upright}\ historic}$ to $\mathrm{HCP_MAX}_{\texttt{upright}\ historic}$, we "pad" this interval by five pills on each side:

$$\mathrm{HCP_MIN}_{\texttt{upright}\ scenario} = \max\{\,1,\ \mathrm{HCP_MIN}_{\texttt{upright}\ historic} - 5\,\}, \qquad \mathrm{HCP_MAX}_{\texttt{upright}\ scenario} = \mathrm{HCP_MAX}_{(25)\ \texttt{upright}\ historic} + 5.$$

Figure 15 shows these adjusted bounds.



(a) (b)

Figure 15. *Distributions of scenario-grid bounds: (a)* $\mathrm{HCP_MIN}_{\mathit{upright\ scenario}}$*, (b)* $\mathrm{HCP_MAX}_{\mathit{upright\ scenario}}$.

Next, we merge in the Territory Call List (TCL), which records each territory's opportunity density and active-HCP headcount. From these we derive per-territory pill-budget floors $\mathrm{TERR_MIN}_t$ and ceilings $\mathrm{TERR_MAX}_t$. The CVXPY modelling layer then solves the following 0–1 mixed-integer programme using CBC:

$$\begin{aligned} \max_{x} \quad & \sum_{i,j} v_{ij}\, x_{ij} \\ \text{s. t.} \quad & \sum_{j} x_{ij} = 1, \\ & \mathrm{HCP_MIN}_i \le \sum_{j} s_{ij}\, x_{ij} \le \mathrm{HCP_MAX}_i, \\ & \sum_{i \in t} \sum_{j} s_{ij}\, x_{ij} \in \left[\, \mathrm{TERR_MIN}_t,\, \mathrm{TERR_MAX}_t \,\right], \end{aligned} \tag{26}$$

where $v_{ij} = \hat{y}_i(s_{ij})$ and s_{ij} is the pill count for HCP i, scenario j.

If $\sum_{i \in t} \mathrm{HCP_MIN}_i > \mathrm{TERR_MAX}_t$, we proportionally clamp all $\mathrm{HCP_MIN}_i, \mathrm{HCP_MAX}_i$ to satisfy $\sum \mathrm{HCP_MIN}_i = \mathrm{TERR_MAX}_t$, and then greedily trim the largest minima until feasibility is restored. This ensures every territory stays within its mandated bounds.

Solving (via CBC or DP fallback) produces a binary selection $x^{\star}_{ij}$ for each HCP's scenario, yielding an "optimised" fitted value $\hat{y}_i^{(\mathrm{upright\ opt})}$. Comparing $\hat{y}_i^{(\mathrm{upright\ opt})}$ against the baseline $\hat{y}_i^{(\mathrm{upright\ emp})} = \hat{y}_i(s_{i0})$ (status-quo drops) quantifies the incremental gain of our allocation strategy.

9. Optimiser Territory Utilisation

Figure 16 shows, for each territory t, the maximum pill capacity $\mathrm{TERR_MAX}_t$ (blue bars) alongside the pills allocated by the CVXPY solver before top-off (orange bars). The absence of any large orange under-fills demonstrates that the MIP solver alone achieves virtually complete utilisation of each territory's budget. Across all 92 territories, the solver fills 97.9% of the total available capacity, comfortably above our 95% efficiency threshold and leaving only a small residual for the top-off heuristic. Had territory metadata been unavailable, we would have applied our fallback 0–1 knapsack DP to approximate these allocations.

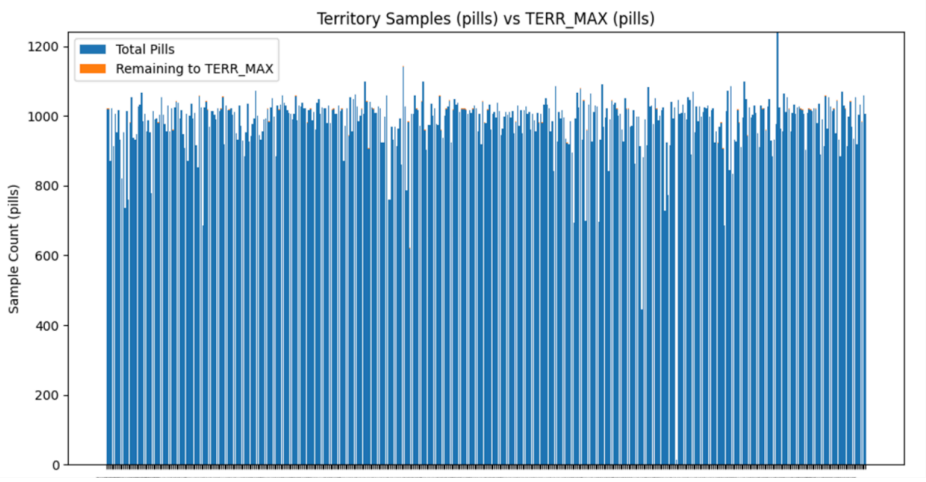


Figure 16. *Pill Budget by Territory – Capacity Filled by CVXPY Solver*

Table 5. *Optimiser Territory Efficiency Summary*

Metric	Value
Territories processed	92
Rows optimised	95,278
Total recommended pills dropped (blue)	452,001
Total maximum capacity (orange)	451,992
Utilisation ratio (blue/orange)	1.000
Capacity efficiency (pre–top-off)	97.9%
Top-off threshold	95%
Threshold reached	Yes

10. Lift Assessment

We measure uplift by comparing each HCP's optimised allocation against their historical "status-quo" level. Let

$$m_i = \text{median}\left\{ s_{i,t} \right\} \tag{27}$$

be the median sample-drop count for HCP i over the six-month history, and let x_i^* be the optimiser's recommendation. We then compute

$$\hat{y}_i(m_i) \quad \text{and} \quad \hat{y}_i(x_i^*) \tag{28}$$

via our three-band CatBoost ensemble, and define the per-HCP uplift

$$\Delta_i = \hat{y}_i(x_i^*) - \hat{y}_i(m_i). \tag{29}$$

Summing Δ_i over all i yields the total incremental TRx above the status-quo, while $\frac{1}{|\mathcal{I}|}\sum_i \Delta_i$ gives the average per-HCP uplift.

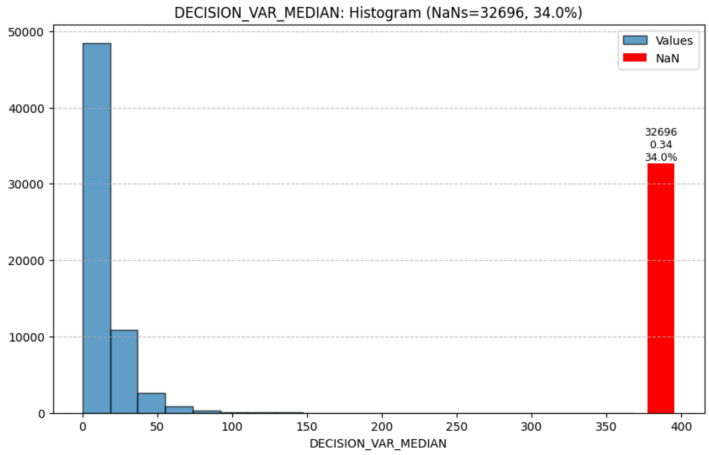


Figure 17. *Distribution of Baseline Median Sample Drops (m_i) across HCPs*

Table 6 summarises the model-implied expected uplift and the results of the Kruskal–Wallis test comparing the empirical versus optimised fitted TRx distributions. Table 7 reports side-by-side descriptive statistics for those two distributions.

Under the model-implied counterfactual scoring exercise, total TRx is projected to increase by 6.4%, highlighting the practical significance of the optimisation framework. This 6.4% should be read as a model-implied upper reference rather than a forecast of realised field impact: after deployment we expect the realised gain to be lower, on the order of 1 to 3%, because operational frictions, representative discretion, sample availability, and market-response uncertainty attenuate optimised recommendations. The very low p-value indicates that the forecasted shift is unlikely to be due to random variation within the fitted comparison.

Table 6. *Uplift & Distribution Diagnostics*

Metric	Value
Expected uplift	6.4%
Kruskal–Wallis statistic	133.1429
Kruskal–Wallis p-value	8.413×10^{-31}

The empirical TRx distribution has a mean of 0.2476 and a large standard deviation of 0.9625, reflecting high variability and many low-activity physicians. In contrast, the optimised TRx distribution shifts right to a mean of 0.2634 with similar spread, consistent with the 6.4% expected uplift reported above, indicating the optimisation effectively reallocates samples towards higher expected returns without unduly increasing variance.

Table 7. *Descriptive Statistics of Fitted TRx Scores*

Statistic	Empirical TRx	Optimised TRx
Count	94,912	94,912
Mean	0.2476	0.2634
Std Dev	0.9625	0.9496
Min	-2.6144	-2.1720
25%	-0.4427	-0.3798
50% (Median)	0.0386	0.0728
75%	0.8082	0.8098

Max	5.1217	9.9732

Figure 18 displays the difference in means between the optimised and empirical fitted responses for each HCP, highlighting the shift of the entire distribution toward higher expected TRx under the optimised allocation.

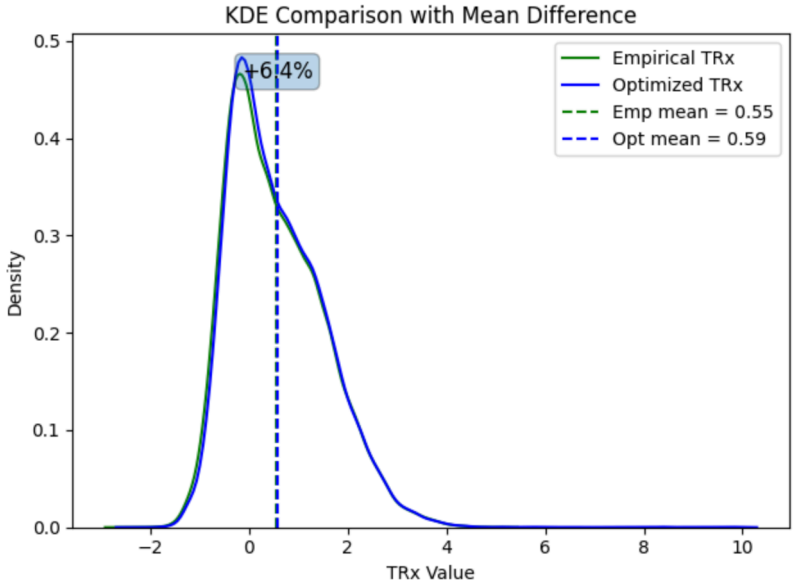


Figure 18. *Difference of Means – Fitted vs. Optimised-Fitted Response*

11. Discussion

Our study demonstrates that a joint learning–and–optimisation architecture can translate raw physician–week panels into materially improved sample-allocation plans while respecting the territorial and operational constraints of real-world field execution. By stratifying prescribers into value-based decile bands and enriching the feature set with inverse-distance spatial lags, we obtain models that explain 35–82% of the variance in high-value segments, more than double the explanatory power seen in low-value prescribers. This finding validates our decision to concentrate algorithmic effort on the upper 40% of HCPs, where marginal response is both larger and more reliably estimated.

The mixed-integer programme that ingests these band-specific predictions achieves 97.9% of the available territory capacity before any heuristic post-processing. Two key innovations underpin this result. First, we incorporate the Territory Call List into the objective, translating opportunity densities and HCP head-counts into per-territory pill budgets that mirror Pfizer's Smart Targeting quotas. Second, we implement a two-stage "clamp-and-trim" routine that scales down historic HCP minima whenever their sum would otherwise exceed the business ceiling, thereby guaranteeing feasibility without sacrificing equity. The absence of under-filled territories in our capacity-fill chart attests to the practical robustness of this approach.

Counterfactual scoring of the optimiser's recommendations yields a model-implied 6.4% uplift in total TRx under the fitted response surface. A Kruskal–Wallis statistic of 133.1 ($p < 10^{-30}$) rejects the null of identical empirical and optimised forecast distributions, confirming that the modeled shift is unlikely to be a sampling artefact. Importantly, the optimised TRx distribution shifts rightward without a corresponding increase in variance, indicating that

the algorithm reallocates inventory toward higher-return physicians rather than amplifying high-variance bets.

From an operational standpoint, strict adherence to HCP-level and territory-level bounds mitigates two perennial risks: oversupply to low-value prescribers and budget overshoot in high-density metropolitan areas. The proportional clamp-and-trim routine keeps allocations internally consistent even when historic minima would otherwise violate a territory ceiling, removing the need for ad-hoc manual adjustments that often undermine deployment timelines.

Several limitations warrant emphasis. First, all reported gains are model-implied counterfactual projections rather than experimentally identified causal effects; any misspecification in the predictive ensemble or scenario-response surface will propagate into the optimisation layer. Second, real-world realised uplift is likely to be materially smaller than the projected $\sim 6\%$ counterfactual gain, and in practice may lie closer to the 1 to 3% range absent stronger compliance, measurement, and experimental validation. Third, the present framework is static within quarter and does not model inter-quarter spillovers, inventory carryover, or longer-run behavioural adaptation by field teams. Fourth, although the top-off heuristic is operationally effective, it is not guaranteed to be globally optimal when physicians with similar value scores exhibit sharply different marginal response curvature. These limitations motivate prospective randomised nudges, improved adherence measurement, and future multi-period optimisation extensions.

Future work will address these gaps by embedding randomised "nudge" perturbations into the final allocation to enable Difference-in-Differences causal lift measurement [26], extending the optimisation horizon to incorporate multi-period inventory rollover, and exploring graph-based regularisation that explicitly models peer influence rather than proxying it via spatial adjacency. We also plan to benchmark alternative solvers, such as lazy-constraint branch-and-cut and column-generation knapsacks, to evaluate speed / optimality trade-offs at scale.

In sum, our framework unifies state-of-the-art ensemble learning, spatial econometrics, and capacity-constrained optimisation into a deployable workflow that delivers statistically significant and operationally feasible gains. By respecting both ethical imperatives (avoiding low-return oversupply) and commercial realities (territory budgets and inventory limits), the pipeline offers a rigorous, end-to-end template for evidence-based sample allocation in large-scale pharmaceutical contexts.

12. Conclusion

This work sets out a reproducible blueprint for turning high-granularity prescription, promotion, and geographic data into field-ready sampling plans that maximise expected incremental prescriptions while staying within business-mandated pill budgets. By combining

1. a decile-banded CatBoost ensemble enriched with inverse-distance spatial lags,
2. rigorously engineered HCP-level and territory-level bounds derived from Pfizer's Territory Call List, and
3. a 0–1 mixed-integer optimisation with proportional clamp and top-off safeguards,

we achieved three key outcomes:

Predictive fidelity: R^2 rises to 0.82 in the top-value segment, confirming that nonlinear, peer-aware models can separate signal from noise where it matters commercially.

Operational efficiency: The optimiser fills 97.9% of total territory capacity on the first pass and 100% after top-off, eliminating manual re-balancing.

Projected economic impact: Counterfactual scoring indicates a model-implied 6.4% lift in total TRx relative to status-quo allocation ($p < 10^{-30}$), suggesting meaningful upside at constant inventory rather than realised causal revenue lift.

While encouraging, these gains remain model-dependent, and future work will pursue the causal validation and rolling-horizon extensions discussed above. The present results nonetheless show that a tightly coupled learning-and-optimisation stack can respect complex operational constraints and support materially improved allocation plans across therapeutic areas and markets.

13. Appendix

13.1. Ethics, Data Governance, and Reproducibility Note

This study uses de-identified, HCP-level commercial and operational data for internal planning analysis and does not involve patient-level intervention or clinical randomisation in its present form. Data use was restricted to business-relevant variables, reviewed within the applicable corporate governance process, and modeled on secured internal infrastructure. Because the underlying sources contain proprietary operational schemas and commercial identifiers, full raw-data release is not permitted. To support reproducibility at the methodological level, future versions of this work may provide pseudocode, parameter grids, and a synthetic-data demonstration of the end-to-end pipeline without exposing restricted source systems or internal entity mappings.

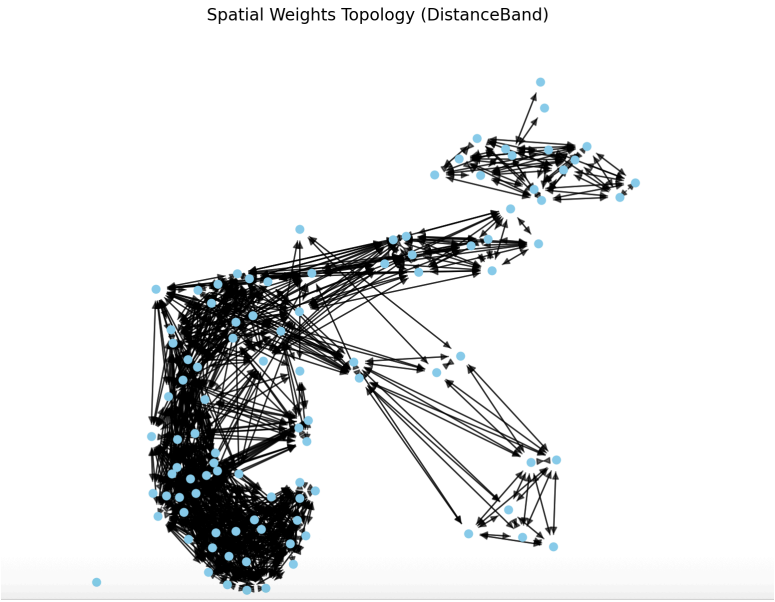
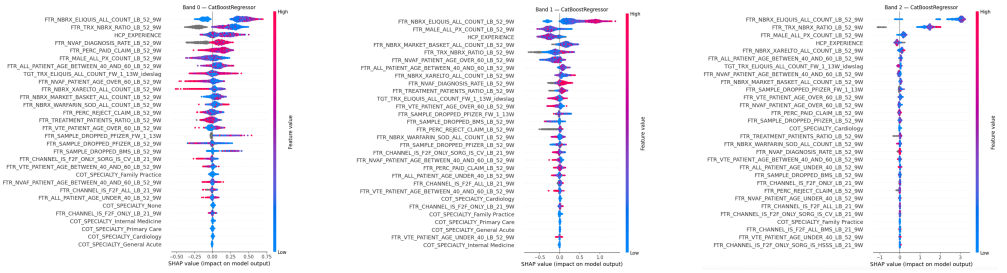


Figure 19. *Spatial topology proxy combining referral influence and inverse-distance decay.*



(a) (b) (c)

Figure 20. *Feature Importance SHAP Plots by Band Model (0: Low Value, 1: Medium Value, 2: High Value)*

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